RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Information Technology
Third Year - Semester II Examination – February - April 2025

ICT 3304 – EMBEDDED SYSTEMS

Time: Three (03) hours

- This paper contains **five (05)** questions in **two (02)** pages.
- Question 01 and 02 are mandatory. Answer any 02 questions from questions 03 to 05.

1. a) Define an embedded system and explain three (03) key characteristics that distinguish it from general-purpose computers. (10 marks)

   b) Describe three (03) real-world applications of embedded systems and explain their impact. (10 marks)

   c) Explain the difference between a microcontroller and a microprocessor with examples. (05 marks)

2. You are tasked with designing an embedded system for an automated smart waste bin. The bin should be able to detect when it is full and send an alert for disposal. Make your own assumptions when needed and state them.

   a) List and explain the key hardware components (sensors, controllers, actuators) required for this system. (10 marks)

   b) Describe how the software logic would work to detect a full bin and trigger an alert. (10 marks)

c) List three (03) challenges in implementing this system in the real world and propose solutions to mitigate them. (05 marks)

3. a) Explain the role of any three (03) of the following components in an embedded system:
Microcontroller
Memory (RAM & ROM)
I/O Interfaces
Sensors
Actuators
Communication Modules (UART, SPI, I2C, etc.)
Power Management Unit (09 marks)

b) Describe synchronous and asynchronous communication with examples. (08 marks)

c) What are polling and interrupt-driven I/O? (04 marks)

d) Explain a scenario where interrupt-driven I/O is preferred. (04 marks)

4. a) Explain why using functions is beneficial in embedded C programming. Provide an example. (10 marks)

b) Draw a flow diagram, write a pseudocode, or write a simple embedded C program to turn ON an LED when a button is pressed. Assume appropriate pin configurations. (10 marks)

c) What is an Interrupt Service Routine (ISR)? (Your explanation should include the purpose of ISR in embedded systems). (05 marks)

5. a) Explain the working principle of hardware timers in embedded systems. Please include how hardware timers can be used for delay generation in your answer. (10 marks)

b) Draw a flow diagram, write a pseudocode, or write an embedded C program to generate a 1-second delay using a timer interrupt (assume an 8-bit timer and a 1MHz clock). (10 marks)

c) What are the key differences between hard real-time and soft real-time systems? Provide examples for each. (05 marks)

--- END ---